

Chapter 5 — Parabolic equations

The method of lines: finite elements in space,
finite differences in time

MTH3340 · Numerical Methods for PDEs

1 A new kind of problem

So far, everything was **steady**: one solve, one function $u(\mathbf{x})$.

Poisson, elasticity $\longrightarrow$ convection–diffusion $\longrightarrow$ now: $u(\mathbf{x}, t)$.

Time-dependent elliptic problems are called **parabolic**. Model: the heat equation.

- the same spatial operator as in Chapters 1–3, plus a time derivative;
- a new datum: the **initial condition**;
- a new question: how to discretise *time*.

This week: *finite elements in space, a time-stepping scheme in time*.

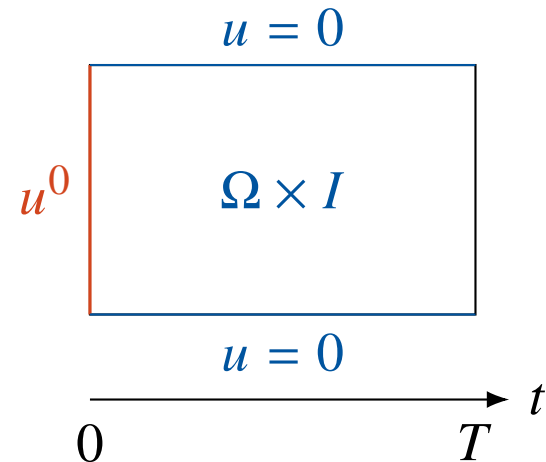
2 The heat equation

$\Omega \subset \mathbb{R}^d$ in space, $I \doteq (0, T]$ in time:

$$\begin{aligned}\partial_t u - \nabla \cdot \mu \nabla u &= f && \text{in } \Omega \times I, \\ u &= g && \text{on } \Gamma_D \times I, \\ \mu \mathbf{n} \cdot \nabla u &= q && \text{on } \Gamma_N \times I, \\ u(\mathbf{x}, 0) &= u^0(\mathbf{x}) && \mathbf{x} \in \Omega.\end{aligned}$$

- u : temperature, μ : conductivity, f : heat source;
- boundary conditions at every time;
- the initial condition u^0 is essential: without it, the problem is not well posed.

As usual: $g = 0$, $\Gamma_D = \partial\Omega$, $\Gamma_N = \emptyset$.



3 Two observations

1. The time derivative is a convective term. In the space-time domain $\Omega \times I \subset \mathbb{R}^{d+1}$,

$$\partial_t u = \beta \cdot \nabla_{(x,t)} u \quad \text{for } \beta = (\mathbf{0}, 1)^T,$$

while diffusion acts *only in space*. Chapter 4 says: **a space-time Galerkin method will not be suitable** (convection with no diffusion along t).

2. **Time causality**. The solution at time t depends on values at $t' < t$ (the past) but not at $t' > t$ (the future).

A standard FEM in space-time ignores this: it **couples all the time DOFs** in one huge system, in both directions.

Method of lines: finite elements in space, finite differences in time.

It respects the *arrow of time*, and that is what makes it affordable.

4 The weak form at each time

Multiply by $v \in H_0^1(\Omega)$, integrate by parts *in space only*:

Definition (Weak heat equation)

Find $u(t) \in H_0^1(\Omega)$ such that, for a.e. $t \in I$,

$$\int_{\Omega} \partial_t u v + \int_{\Omega} \mu \nabla u \cdot \nabla v = \int_{\Omega} f v \quad \forall v \in H_0^1(\Omega).$$

Functions of time with values in a space V ($V = L^2(\Omega)$ or $H^1(\Omega)$):

$$\|u\|_{L^2(I;V)}^2 \doteq \int_0^T \|u(t)\|_V^2 dt, \quad \|u\|_{C^0(I;V)} \doteq \sup_{t \in I} \|u(t)\|_V.$$

Where does the solution live?

$$u \in C^0(I; L^2(\Omega)) \cap L^2(I; H^1(\Omega)) :$$

continuous in time in the L^2 norm, and with finite *total* energy dissipation.

We will prove exactly this at the discrete level.

5 The method of lines

Same $V_h = \text{span}\{\varphi^1, \dots, \varphi^N\}$, but the **DOFs depend on time**, and Galerkin holds at every $t \in I$:

$$u_h(\mathbf{x}, t) \doteq \sum_{a=1}^N \varphi^a(\mathbf{x}) U^a(t), \quad \int_{\Omega} \partial_t u_h v_h + \int_{\Omega} \mu \nabla u_h \cdot \nabla v_h = \int_{\Omega} f v_h.$$

Semi-discrete: discrete in space, continuous in time. Take $v_h = \varphi^a$ and expand u_h :

$$\underbrace{\left[\int_{\Omega} \varphi^a \varphi^b \right]}_{M_{ab} \text{ (mass)}} \dot{U}^b + \underbrace{\left[\int_{\Omega} \mu \nabla \varphi^a \cdot \nabla \varphi^b \right]}_{K_{ab} \text{ (stiffness)}} U^b = \underbrace{\int_{\Omega} f \varphi^a}_{F_a} \quad a = 1, \dots, N.$$

$$M\dot{U} + KU = F \quad \text{for } t \in I, \quad U(0) = \text{DOFs of } u^0 :$$

a **system of N linear ODEs**. Since M is SPD, $\dot{U} = M^{-1}(F - KU)$, and the Picard–Lindelöf theorem gives **existence and uniqueness** of a solution $U \in C^1(I)$ (MTH2051).

6 Stability of the semi-discrete problem

Theorem (Stability of the semi-discrete heat equation)

For every $s \in I$,

$$\|u_h(s)\|_{L^2(\Omega)}^2 + \int_0^s \mu \|\nabla u_h\|_{L^2(\Omega)}^2 dt \leq \|u_h(0)\|_{L^2(\Omega)}^2 + \int_0^s \frac{C_P^2}{\mu} \|f(t)\|_{H^{-1}(\Omega)}^2 dt.$$

Take $v_h = u_h$. The diffusive and forcing terms are handled as in the steady case, $\int_{\Omega} f u_h \leq \frac{C_P^2}{2\mu} \|f\|_{H^{-1}}^2 + \frac{\mu}{2} \|\nabla u_h\|^2$. The new term is the time derivative:

$$\int_{\Omega} u_h \partial_t u_h = \frac{1}{2} \int_{\Omega} \partial_t (u_h^2) = \frac{1}{2} \frac{d}{dt} \|u_h\|_{L^2(\Omega)}^2.$$

Hence $\frac{1}{2} \frac{d}{dt} \|u_h\|^2 + \frac{\mu}{2} \|\nabla u_h\|^2 \leq \frac{C_P^2}{2\mu} \|f\|_{H^{-1}}^2$; integrate from 0 to s and multiply by 2. $\square$

L^2 in space uniformly in time, $u_h \in C^0(I; L^2(\Omega))$; H^1 in space integrated in time, $u_h \in L^2(I; H^1(\Omega))$: finite total dissipation.

7 Convergence in space: the Ritz projector

Expectation: h^{r+1} in L^2 , h^r in H^1 . **Obstacle:** splitting through the interpolant leaves the diffusive term $\mu(\nabla(u - I_h u), \nabla v_h)$ in the error equation, worth only h^r : it pollutes the L^2 estimate.

Cure: at each t , compare with the **Ritz projector**,

$$\mathcal{R}_h u \in V_h : \quad \int_{\Omega} \nabla(u - \mathcal{R}_h u) \cdot \nabla v_h = 0 \quad \forall v_h \in V_h,$$

a Poisson problem at each t (Chapters 2–3; the L^2 bound by duality):

$$\|u - \mathcal{R}_h u\|_{L^2} + h \|u - \mathcal{R}_h u\|_{H^1} \leq C h^{r+1} |u|_{H^{r+1}}, \quad \partial_t(\mathcal{R}_h u) = \mathcal{R}_h(\partial_t u).$$

Set $e_h \doteq \mathcal{R}_h u - u_h \in V_h$ and $\rho \doteq \mathcal{R}_h u - u$. Galerkin orthogonality and the definition of $\mathcal{R}_h$:

$$(\partial_t e_h, v_h) + \underbrace{\mu(\nabla e_h, \nabla v_h)}_{=0} = (\partial_t \rho, v_h) \quad \forall v_h \in V_h.$$

$v_h = e_h$, as in the stability proof:

$$\frac{1}{2} \frac{d}{dt} \|e_h\|_{L^2}^2 + \mu \|\nabla e_h\|_{L^2}^2 \leq (\partial_t \rho, e_h). \quad (*)$$

Two estimates from (*): L^2 first, then H^1 .

8 (1) The L^2 error

Gronwall: $\phi' \leq \beta\phi + g$ on $I \implies \phi(t) \leq e^{\beta t}\phi(0) + \int_0^t e^{\beta(t-s)}g(s) \, ds$.

Young on the right of (*) with weight μ/C_P^2 , Poincaré on the left:

$$\frac{d}{dt}\|e_h\|^2 + \frac{\mu}{C_P^2}\|e_h\|^2 \leq \frac{C_P^2}{\mu}\|\partial_t \rho\|^2 \quad (\text{norms in } L^2(\Omega)).$$

Gronwall with $\beta = -\mu/C_P^2$, the maximum of $\|\partial_t \rho\|^2$ out of the integral, and $\int_0^t e^{\beta(t-s)}ds \leq C_P^2/\mu$:

$$\|e_h(t)\|^2 \leq e^{-\mu t/C_P^2}\|e_h(0)\|^2 + \frac{C_P^4}{\mu^2} \max_{s \leq t} \|\partial_t \rho(s)\|^2.$$

Square roots, $\|u - u_h\| \leq \|\rho\| + \|e_h\|$, $e^{-\mu t/C_P^2} \leq 1$, and the Ritz bounds:

$$\begin{aligned} \|u - u_h\|_{C^0(I;L^2(\Omega))} \leq C \Big[& h^{r+1}|u(0)|_{H^{r+1}} + h^{r+1}\|u\|_{C^0(I;H^{r+1})} \\ & + \frac{C_P^2}{\mu} h^{r+1}\|u_t\|_{C^0(I;H^{r+1})} \Big]. \end{aligned}$$

9 (2) The H^1 error

Back to (*): H^{-1} – H^1 duality and Young with a free $\alpha > 0$,

$$(\partial_t \rho, e_h) \leq \frac{1}{2\mu\alpha} \|\partial_t \rho\|_{H^{-1}}^2 + \frac{\mu\alpha}{2} \|e_h\|_{H^1}^2;$$

Poincaré $\|e_h\|_{H^1} \leq C_P |e_h|_{H^1}$, and α small enough to absorb:

$$\frac{d}{dt} \|e_h\|_{L^2}^2 + \mu |e_h|_{H^1}^2 \leq \frac{C}{\mu} \|\partial_t \rho\|_{H^{-1}}^2.$$

Integrate from 0 to t — no Gronwall needed:

$$\mu \int_0^t |e_h|_{H^1}^2 \leq \|e_h(0)\|_{L^2}^2 + \frac{C}{\mu} \int_0^t \|\partial_t \rho\|_{H^{-1}}^2.$$

Then $\|\partial_t \rho\|_{H^{-1}} \lesssim \|\partial_t \rho\|_{L^2} \lesssim h^{r+1}$, integrate over I , square roots, $|\rho|_{H^1} \lesssim h^r$:

Theorem (Error estimates for the semi-discrete heat equation)

$$\begin{aligned} \mu^{1/2} \|u - u_h\|_{L^2(I; H^1(\Omega))} \leq C \Big[& h^{r+1} |u(0)|_{H^{r+1}} + \sqrt{\frac{T}{\mu}} h^{r+1} \|u_t\|_{C^0(I; H^{r+1})} \\ & + \sqrt{T} \mu^{1/2} h^r \|u\|_{C^0(I; H^{r+1})} \Big]. \end{aligned}$$

10 Time discretisation: the θ -method

A **finite difference** method in time. Partition $I = [0, T]$, and let $U^i \approx U(t_i)$:



Assume U linear on $[t_i, t_{i+1}]$ and impose the ODE at $t_i + \theta\tau_i$:

$$\dot{U} \approx \frac{U^{i+1} - U^i}{\tau_i}, \quad U(t_i + \theta\tau_i) \approx U^{i+\theta} \doteq \theta U^{i+1} + (1 - \theta)U^i.$$

Definition (θ -method, $\theta \in [0, 1]$)

U^0 given. For $i = 0, \dots, M - 1$, find U^{i+1} from

$$M \frac{U^{i+1} - U^i}{\tau_i} + K U^{i+\theta} = F^{i+\theta}.$$

Time marching: causality exploited, one time level at a time.

Three members of the family

$$\theta = 1 \quad \text{backward Euler (BE)} \quad (M + \tau_i K) U^{i+1} = \tau_i F^{i+1} + M U^i$$

$$\theta = 0 \quad \text{forward Euler (FwE)} \quad M U^{i+1} = \tau_i F^i + (M - \tau_i K) U^i$$

$$\theta = \frac{1}{2} \quad \text{Crank–Nicolson (CN)} \quad (M + \frac{\tau_i}{2} K) U^{i+1} = \tau_i F^{i+\frac{1}{2}} + (M - \frac{\tau_i}{2} K) U^i$$

Implicit (BE, CN): solve with $M + c\tau K$

- symmetric positive definite:
 $U^T M U = \|u_h\|_{L^2}^2, \quad U^T K U = \mu \|\nabla u_h\|_{L^2}^2;$
- a Poisson-like solve *per step*;
- **unconditionally stable**.

Explicit (FwE): solve with M only

- mass matrix: very cheap (spectrally an identity);
- **conditionally stable**: a CFL restriction on τ ;
- the price of cheap steps.

Accuracy in time (MTH2051): BE and FwE first order, $O(\tau)$; CN second order, $O(\tau^2)$. **Now the three stability claims.**

11 Stability of backward Euler

Uniform τ . Premultiply $M(U^{i+1} - U^i) + \tau KU^{i+1} = \tau F^{i+1}$ by $(U^{i+1})^T$.
 Notation $\|U\|_X^2 \doteq U^T X U$.

The trick, for the time-derivative term:

$$(a - b) a = \frac{1}{2}(a^2 - b^2 + (a - b)^2) \implies$$

$$(U^{i+1})^T M(U^{i+1} - U^i) = \frac{1}{2}\|U^{i+1}\|_M^2 - \frac{1}{2}\|U^i\|_M^2 + \frac{1}{2}\|U^{i+1} - U^i\|_M^2.$$

The forcing term as always: $\tau(U^{i+1})^T F^{i+1} \leq \frac{\tau}{2}\|F^{i+1}\|_{K^{-1}}^2 + \frac{\tau}{2}\|U^{i+1}\|_K^2$. Then

$$\frac{1}{2}\|U^{i+1}\|_M^2 - \frac{1}{2}\|U^i\|_M^2 + \frac{1}{2}\|U^{i+1} - U^i\|_M^2 + \frac{\tau}{2}\|U^{i+1}\|_K^2 \leq \frac{\tau}{2}\|F^{i+1}\|_{K^{-1}}^2.$$

Sum over $i = 0, \dots, m - 1$: the first two terms telescope.

Lemma (Stability of BE)

$$\|U^m\|_M^2 + \sum_{i=1}^m \|U^i - U^{i-1}\|_M^2 + \sum_{i=1}^m \tau \|U^i\|_K^2 \leq \|U^0\|_M^2 + \sum_{i=1}^m \tau \|F^i\|_{K^{-1}}^2.$$

Reading the bound

$$\underbrace{\|U^m\|_M^2}_{\|u_h^m\|_{L^2}^2} + \underbrace{\sum_{i=1}^m \tau \|U^i\|_K^2}_{\approx \int_0^{t_m} \mu \|\nabla u_h\|^2} + \underbrace{\sum_{i=1}^m \|U^i - U^{i-1}\|_M^2}_{\text{new}} \leq \dots$$

- the first two terms are the **semi-discrete stability**, evaluated at the time nodes;
- the third is **not in the continuous problem**: it is **numerical dissipation** introduced by the scheme.

BE damps every jump $U^i - U^{i-1}$: it is *unphysically* dissipating energy. Robust, but it smooths more than the equation does.

Stability holds for every $\tau > 0$: unconditional, at the cost of one implicit solve per step.

12 Stability of Crank–Nicolson

$$M(U^{i+1} - U^i) + \tau K U^{i+\frac{1}{2}} = \tau F^{i+\frac{1}{2}}, \text{ with } U^{i+\frac{1}{2}} \doteq \frac{1}{2}(U^{i+1} + U^i).$$

Premultiply by $(U^{i+\frac{1}{2}})^T$, not by $(U^{i+1})^T$. **The other trick:**

$$\frac{1}{2}(a+b)(a-b) = \frac{1}{2}a^2 - \frac{1}{2}b^2 \implies (U^{i+\frac{1}{2}})^T M(U^{i+1} - U^i) = \frac{1}{2}\|U^{i+1}\|_M^2 - \frac{1}{2}\|U^i\|_M^2.$$

No square of the jump appears. Forcing as before, sum, telescope:

Lemma (Stability of CN)

$$\|U^m\|_M^2 + \sum_{i=1}^m \tau \|U^{i-\frac{1}{2}}\|_K^2 \leq \|U^0\|_M^2 + \sum_{i=1}^m \tau \|F^{i-\frac{1}{2}}\|_{K^{-1}}^2.$$

- **no numerical dissipation:** exactly the semi-discrete bound, nothing added;
- unconditionally stable, and **second order** in time;
- the price: rough data are *not* damped — see below.

13 Stability of forward Euler: the CFL condition

$M(U^{i+1} - U^i) + \tau K U^i = \tau F^i$. Premultiply by $(U^{i+1})^T$ as for BE: now $(U^{i+1})^T K U^i$ appears instead of $\|U^{i+1}\|_K^2$, and it **has no sign**:

$$\begin{aligned}\tau(U^{i+1})^T K U^i &= \tau\|U^{i+1}\|_K^2 - \tau(U^{i+1})^T K (U^{i+1} - U^i) \\ &\geq \frac{\tau}{2}\|U^{i+1}\|_K^2 - \frac{\tau}{2}\|U^{i+1} - U^i\|_K^2.\end{aligned}$$

The red term must be absorbed by the Euler dissipation $\frac{1}{2}\|U^{i+1} - U^i\|_M^2$; the inverse inequality (Week 8) compares the norms:

$$\tau\|V\|_K^2 = \tau\mu\|\nabla v_h\|_{L^2}^2 \leq C_{\text{inv}} \frac{\tau\mu}{h^2} \|v_h\|_{L^2}^2 = C_{\text{inv}} \frac{\tau\mu}{h^2} \|V\|_M^2.$$

Lemma (Stability of FwE)

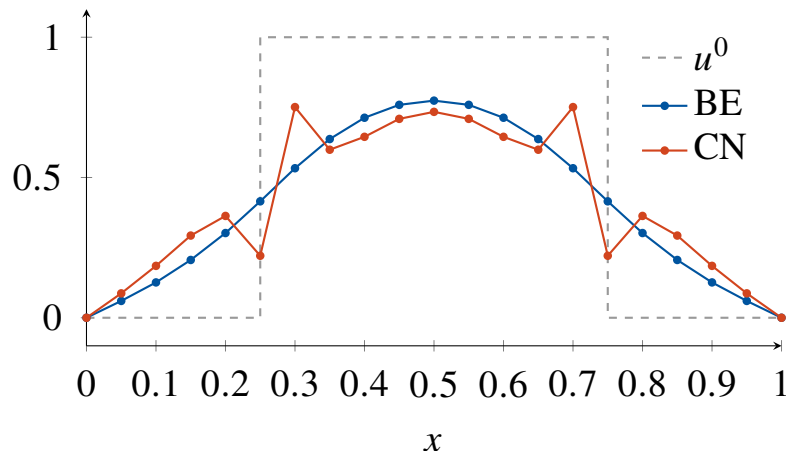
*Under the **CFL condition** $\tau < h^2 / (C_{\text{inv}} \mu)$, the BE bound holds, with a constant in front of the dissipation term.*

$\tau \sim h^2$: halve h , quarter τ . Convection has a milder CFL, $\tau \sim h$: hence **semi-implicit** schemes (explicit convection, implicit diffusion).

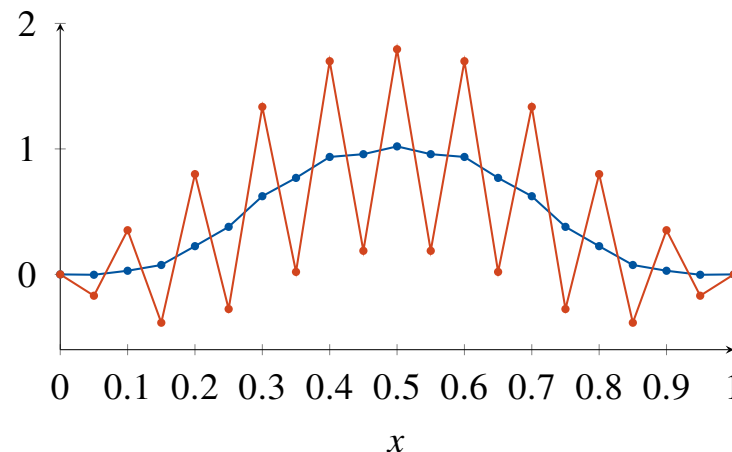
Seeing it in 1D

$$\partial_t u - u'' = 0 \text{ on } (0, 1), u(0, t) = u(1, t) = 0, u^0 = \chi_{(0.25, 0.75)}, \mathcal{P}_1, h = 1/20.$$

implicit, $\tau = 10^{-2}$, $t = 0.02$



forward Euler, $t = 0.004$: $\tau = 4 \cdot 10^{-4}$, $\tau = 5 \cdot 10^{-4}$



- **BE**: jumps gone after two steps (numerical dissipation); **CN**: same τ , stable, but the jumps *ring* — nothing damps them;
- **FwE**: CFL limit $h^2/6 \approx 4.2 \cdot 10^{-4}$; just above it, node-to-node oscillations grow every step ($\max |U| \approx 35$ at $t = 0.01$).

14 Total error

Two sources of error, one from each discretisation:

$$u(t_i) - u_h^i = \underbrace{(u(t_i) - u_h(t_i))}_{\text{space: } h^{r+1}, h^r} + \underbrace{(u_h(t_i) - u_h^i)}_{\text{time: } \tau^\beta}.$$

Theorem (Error estimates for the fully discrete heat equation)

For u smooth enough in space and time, a space of order r , and $\theta \in \{1, \frac{1}{2}\}$,

$$\|u(t_i) - u_h^i\|_{L^2(\Omega)} \leq C(h^{r+1} + \tau^\beta), \quad i = 1, \dots, M,$$

$$\left(\sum_{n=1}^M \tau \mu \|u(t_n) - u_h^n\|_{H^1(\Omega)}^2 \right)^{1/2} \leq C(h^r + \tau^\beta), \quad \beta = \begin{cases} 1 & \theta = 1 \text{ (BE)}, \\ 2 & \theta = \frac{1}{2} \text{ (CN)}. \end{cases}$$

Balancing the two. Refining h alone is pointless once τ^β dominates:

$$\text{BE: } \tau \sim h^{r+1} \text{ (for the } L^2 \text{ error),} \quad \text{CN: } \tau \sim h^{(r+1)/2}.$$

$\mathcal{P}_1$, L^2 error: BE needs $\tau \sim h^2$, CN only $\tau \sim h$.

15 Summary

	solve	stability	order	dissipation
BE ($\theta = 1$)	$M + \tau K$	unconditional	τ	numerical
CN ($\theta = \frac{1}{2}$)	$M + \frac{\tau}{2}K$	unconditional	τ^2	none
FwE ($\theta = 0$)	M	$\tau < h^2/(C_{\text{inv}}\mu)$	τ	if stable

The method of lines.

- space: Galerkin FEM at every $t \Rightarrow$ ODE system $M\dot{U} + KU = F$;
- semi-discrete analysis: the same energy argument plus $\frac{1}{2}\frac{d}{dt}\|u_h\|^2$; Ritz projector for optimal L^2 rates;
- time: θ -method; stability by premultiplying with the right vector and one algebraic identity; total error $h^{r+1} + \tau^\beta$.

Causality is the design principle: nothing couples a time level with its future.